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FAST Version 2.4.4
User's Manual

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This documentation and the software described in this documentation are subject to change without prior notice.

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1. INTRODUCTION

FAST (from *Finite element AnalysiS Tool*) is a program for structural analysis using the Finite Element Method (FEM) and Isogeometric Analysis (IGA). FAST is developed at Laboratório de Mecânica Computacional e Visualização (LMCV) of the Universidade Federal do Ceará (UFC).

This manual describes the main features of the FAST program. It is assumed that the reader is familiar with the Finite Element Method (FEM) for linear and nonlinear structural as described in reference books (Hughes, 1987; Crisfield, 1991; Bathe, 1996; Crisfield, 1997; Belytschko et al., 2000; Cook et al., 2002; Zienkiewicz and Taylor, 2005) and intends to use the program to solve a specific problem.

It is important to note that FAST was developed for teaching and research purposes. Therefore, contrary to commercial software, it may contain not only good modeling options, but also some poor ones, such as elements that suffer from locking and integration orders leading to spurious modes. It is up to the user to choose the best modeling options for the problem at hand.

FAST has no built-in units. Therefore, the user should use consistent units for the input data. The input data of the finite element model is carried-out is read from a file with extension .dat (e.g. filename.dat) and the computed results are written in an output file with extension .pos (e.g. filename.pos). Both input and output are text files with a simple structure and can be easily read and written by pre and post-processors. Additionally, they can be opened, written or modified using standard text editors. The input and output files follow the general rules of the neutral file format (https://web.tecgraf.puc-rio.br/neutralfile/), but may use different keywords.

1.1. History

FAST is written in C++ using the Object-Oriented Programming (OOP) paradigm. It is a direct descendent of the FEMOOP program developed at PUC-Rio (Martha et al., 1996; Martha and Parente, 2002).

Since FEMOOP is a very large and complex program, when I began teaching a nonlinear analysis course in 2000, I implemented a *Simple Non-Linear Analysis Program* (SNLAP) to be used by the students as a platform for implementation of finite elements and nonlinear algorithms. In 2005 this program was used as a base for the development of the

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CAP3D program for finite element analysis of pavement structures carried out in collaboration with the team of Laboratório de Mecânica dos Pavimentos (LMP) of the Universidade Federal do Ceará (Holanda et al., 2006; Araújo *et al*., 2010).

The FAST development began in 2011, aiming to build a computational platform for research in analysis and design of structures made of composite materials, include reinforced concrete and fiber reinforced polymers. The program development is carried out to give great flexibility to implementation of complex elements, materials and algorithms, while avoiding unnecessary overhead for more conventional analyses, thus *keeping FAST fast*. This is an important feature, since a major FAST application is in the structural optimization using bio-inspired algorithms, which requires a large number of analyses to solve the optimization problem.

Many former students contributed to the FAST's development and their contribution is gratefully acknowledged. Marcelo Meireles implemented the material nonlinear analysis of frames by the fiber method (Meireles, 2012; Parente *et al*., 2014). Iuri Rocha implemented most of the code for geometrical and material nonlinear analysis of laminated plates and shallow shells (Rocha, 2013). Luiz Antonio Mororó implemented 3D frame elements for large displacement and rotation analysis (Mororó, 2013; Mororó *et al*., 2015; Mororó *et al*., 2020). Edson Dantas implemented the laminated solid elements, interface elements and cohesive zone models (Dantas, 2014). Carlos David Melo implemented the 1D plasticity models (Melo, 2015). Pedro Sanderson Barros implemented progressive failure models for laminates (Barros, 2016). Bergson Matias implemented the HHT and Generalized-α methods and Mazars and Lee-Fenves 1D concrete models (Matias, 2018). Renan Barros implemented the J2 plasticity (Barros, 2023).

Elias Barroso implemented the isogeometric analysis of solids based on NURBS (Barroso, 2015) and the isogeometric analysis of shells based Bézier triangles and quadrilaterals (Barroso, 2022). Jamires Praciano and Samir Auad extended the IGA implementation to laminated and functionally graded (FG) shallow shells (Praciano, 2018; Praciano *et al*., 2019; Auad *et al*., 2019) and cylindrical shells (Auad, 2019). Marina Maia implemented the vibration analysis of FG shallow shells (Maia, 2020; Maia *et al*., 2021). Francisco Davyd Silva implemented the analysis of FG shallow shells under thermal loads using Bézier triangles (Silva, 2021; Silva *et al*., 2024). Pedro Ygor Mesquista and Renan Barros implemented the higher-order plate elements (Barros *et al*., 2023). Matheus Sousa implemented the isogeometric elements for geometrically nonlinear analysis of general shells (Sousa, 2024).

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2. INPUT FILE

The general structure of the input file (.dat) is composed of %KEYWORDS followed by the associated data. In this manual, integer and real values will be indicated as [data] and <data>, respectively. Strings should be written in the input file between ' ' and will be indicated in the manual as 'data(size)', where size (optional) is the maximum size of the string.

The input file is divided in four sections written in the following order: header, nodal data, element data and loading. Thus, general format of the input file is:

```
HEADER

%REMARK
This label can appear anywhere in the file (between sections) and its contents are ignored. It can be
used to document the file.

NODAL DATA

PATCH DATA

ELEMENT DATA

SPRINGS

LOADING

%END
```

The end of the input file should be indicated by the keyword %END. All data after the %END keyword will not the read by the program.

The keywords and associated data of each section will be presented in the following. It is important to note that the keywords should appear in the file in the same order that they appear in this manual.

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2.1. Header

The HEADER section contains general information about the input file and the problem to be solved, as well as the analysis type and algorithm used in analysis. The current available algorithms are:

- Static Analysis:
  - Linear (*default option*): 'Linear' or ‘linear’;
  - Load Control with Newton-Rapshon iterations: 'LoadControl', ‘lcm’, 'NewtonRaphson' or ‘snr’;
  - Displacement Control: 'DisplacementControl' or ‘dcm’;
  - Riks Arc-Length (orthogonal to initial plane): 'InitialOrthoArcLen' or ‘ioal’;
  - Ramm Arc-Length (orthogonal planes): 'UpdatedOrthoArcLen' or ‘uoal’;
  - Consistently Linearized Arc-Length: 'LinearizedArcLen' or ‘clinal’;
  - Crisfield Arc-Length (quadratic): 'CylindricalArcLen' or ‘clal.
- Time-Dependent:
  - Linear quasi-static (viscoelastic): 'QuasiStatic';
  - Linear dynamic: 'Newmark' or 'newmark';
  - Linear dynamic using Hilber-Hughes-Taylor method: 'HHT' or 'hht';
  - Linear dynamic using Generalized-α method: ‘GenAlpha' or 'genalpha';
  - Dynamic/Viscoelastic: 'NewmarkIncremental';
  - Nonlinear dynamic: 'NewmarkNonlinear' or 'newmarknl';
  - Nonlinear dynamic using Hilber-Hughes-Taylor method: 'HHTNonlinear' or 'hhtnl';
  - Nonlinear dynamic using Generalized-α method: ‘GenAlphaNonlinea' or 'genalphanl'.
- Eigenvalue analysis:
  - Vibration frequencies/modes: 'ModalAnalysis';
  - Buckling: 'LinearStability'.

The global matrices (e.g. stiffness and mass) can be stored in three different formats:

- Symmetric matrix in skyline storage (default option);
- Unsymmetric matrix in skyline storage;

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- Sparse symmetric matrix;
- Unsymmetric sparse matrix.

The skyline formats are used with direct solvers based on **LDL**^{T} and **LDU** decompositions for symmetric and unsymmetric matrices, respectively. The sparse format is used with a conjugate gradient solver, which in the current version can be used only for linear static analysis. Unsymmetric sparse matrix implementations from Eigen library is also available, where a conjugated gradient solver with Incomplete Cholesky preconditioner is used (Guennebaud et al., 2010). This option exploits shared memory parallelization via OpenMP, thus it is recommended to obtain better performance in case of problems with large symmetric systems.

The keywords and associated data are:

%HEADER

This section should be in the beginning of the file and should contain any relevant information about the current file.

%HEADER.ANALYSIS.PROGRAM

'FAST'

Note: This item is ignored by FAST, but can be useful for post-processors, since several programs use the neutral file format, but each one of these programs can use different keywords and data. Thus, it is interesting to indicate for which analysis program this input file was created.

%HEADER.ANALYSIS.ALGORITHM

'algorithm' (default value: 'Linear')

%HEADER.ANALYSIS.ARC-LENGTH

<load_coef>

Note: Load coefficient for spherical arc-length (default 0.0, i.e. cylindrical arc-length).

%HEADER.ANALYSIS.DISPLACEMENT.CONTROL

[id_control_node] [controlled_dof (1 to 6)]

Note: Required by the Displacement Control Method.

%HEADER.ANALYSIS.EIGEN.ALGORITHM

'eigen_algorithm_type' (default value: 'InversePower')

Note: The options are: 'InversePower' and 'GeneralizedJacobi'. In case of 'InversePower', only the first eigen pair is computed.

%HEADER.ANALYSIS.EIGEN.NUMBER.MODES

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[#_of_eigen_modes] (default value: 5)

%HEADER.ANALYSIS.GEN-ALPHA.PARAMETERS

< α_m> < α_f>

Note: The default values are $\alpha_m = \alpha_f = 0.0$ (trapezoidal rule without numerical damping).

%HEADER.ANALYSIS.HHT.PARAMETER

<α>

Note: This parameter ($-0.5 \leq \alpha \leq 0.0$ controls the numerical damping. The default value is $\alpha = 0.0$ (trapezoidal rule without numerical damping).

%HEADER.ANALYSIS.MATRIX.TYPE

'matrix_type' (default value: 'skyline')

Note: The options are: 'skyline', 'unsym_skyline', 'unsym_sparse' and 'eigen_unsym_sparse'.

%HEADER.ANALYSIS.NEWMARK.PARAMETERS

<γ> <β>

Note: The default values are $\gamma = 0.5$ and $\beta = 0.25$ (trapezoidal rule).

%HEADER.ANALYSIS.MAXIMUM.ITERATIONS

[max_#_of_iterations_per_step]

%HEADER.ANALYSIS.MAXIMUM.STEPS

[max_step]

%HEADER.ANALYSIS.PRINT.STEPS

[#_of_steps_between_printing]

%HEADER.ANALYSIS.PROPORTIONAL.LOADING

[proportional_loading] (0 = false, 1 = true)

Note: The default value is true for static analysis and false for dynamic analysis. The use of non-proportional loading allows the consideration of different time functions for each external load.

%HEADER.ANALYSIS.RAYLEIGH.DAMPING

<ω_1> < ω_2> <ξ_1> < ξ_2>

Note: The default is no Rayleigh damping ($\xi_1 = \xi_2 = 0$). In the current version this option is available only for Newmark algorithms (linear/nonlinear).

%HEADER.ANALYSIS.STEP.FACTOR

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<incremental_step_factor>

Note: This value corresponds to the incremental load factor for Load Control Method, to the initial load factor for Arc-Length Methods and for the displacement increment for Displacement Control Method.

%HEADER.ANALYSIS.TARGET.ITERATIONS

[#_of_desired_iterations_per_step]

Note: Valid only for Arc-Length Methods.

%HEADER.ANALYSIS.TIME.STEPS

<time_step>

%HEADER.ANALYSIS.TOLERANCE

<tolerance_for_convergence>

%HEADER.PRINT.EQUIVALENT.MESH

[equivalent_mesh_flag] (0 = false, 1 = true)

Note: This flag is used to create an equivalent finite element mesh for post-processing (nodal scalar data). Default value is false. The equivalent mesh is written in a file named [Input .dat name]_EqvMsh.pos.

%HEADER.EQUIVALENT.MESH.DIVISION

[div_r] [div_s] [div_t] (default value is 0)

Note: This flag controls the element discretization used in equivalent mesh.

%HEADER.PRINT.SUPPORT.REACTIONS

[support_reactions_flag] (0 = false, 1 = true)

%HEADER.OUTPUT.STRESS.EXTRAPOLATION

[stress_extrapolation_flag] (0 = false, 1 = true)

Note: This flag control whether nodal stresses are computed by extrapolating results at integration points. Default value is true. It is not recommended in high-order isogeometric models.

%HEADER.OUTPUT.PRECISION

<decimal_precision> (default: 6)

%HEADER.OUTPUT.PATCH.STRESS.SMOOTH

[patch_stress_smooth] (0 = false, 1 = true)

Note: This flag control whether stresses evaluated at patch points must be smoothed. It happens when patch point is located at patch knot spans. Default value is true.

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2.2. Nodal Data

This section describes the nodal data, including coordinates, support conditions, masses, etc. Nodes are defined by their numerical "id" and should be defined sequentially in the ascending order (1, 2, 3, …). The keywords and associated data are:

%NODE
[#_of_nodes]

%NODE.COORD
[#_of_nodal_coordinates]
[node_id] <x><y><z>

%NODE.SUPPORT
[#_of_restrained_node]
[node_id] [dx] [dy] [dz] [rx] [ry] [rz] (0 = free, 1 = fixed)

%NODAL.MASS
[#_of_nodal_masses]
[node_id] <mx><my><mz><ix><iy><iz>

%NODE.CONSTRAINT
[#_of_constraint_nodes]
[constrained_node] [free_node] [dx] [dy] [dz] [rx] [ry] [rz]
Note: constrained_node has the same dof as the free node if the dof code is equal 1. Otherwise, they have different DOFs.

%NODE.SOLVER.ORDER
[#_of_nodes]
[node_id_first] …. [node_id_last]
Note: This item should be used to change the order used to define the nodal DOFs aiming to minimize the profile (skyline) of the global matrices.

%CONTROL.POINT.WEIGHT
[#_of_nodal_coordinates]
[node_id] <w>

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2.3. Patch Data

This section describes the patch data, a geometry entity used in isogeometric elements:

```
%PATCH
[#_of_patches]
[patch_id]  [rational_flag] (0:false,  1: true)  [patch_type]   [patch_data]
Note: [patch_type] can be 'bspsurf' or 'bspsol'. In both cases, [patch_data] consists in:
  For each patch dimension:
     [degree]  [knot_vector]
  [control_point_first] ... [control_point_last].

Knot vectors are read according with given input rule:
'knot_vector_input_rule' possible values are: 'General' and 'Uniform'.

In 'General' case:
[#_of_knots]  [knot_value_first] ... [knot_value_last]  [knot_multiplicity_first] … [knot_multiplicity_last]

In 'Uniform' case:
[#_of_knots]  [knot_value_first]  [delta_value]  [knot_multiplicity_first] … [knot_multiplicity_last]

It is important to note that open knot vectors are considered in Isogeometric formulation.

%PATCH.OUTPUT.POINT
[#_of_patch_output_point]
[patch_id]   <r><s><t>
```

2.4. Element Data

This section describes the element data, including materials, cross sections, element type, section and nodes.

2.4.1. Materials

This part of the input data defines the Material types used in the finite element model. Each material has a "type", "id", and "properties". The id number should be defined sequentially on ascending order (1, 2, 3, …). The mechanical properties depend on the chosen material type. The material types available are:

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- Elastic Isotropic (MATERIAL.ISOTROPIC) – linear elastic material;
- Elastic Orthotropic (MATERIAL.ORTHOTROPIC) – linear elastic material;
- Nonlinear elastic material (MATERIAL.NONLINEAR.ELASTIC) – piecewise linear 1D stress-strain curve;
- Concrete materials with 1D stress-strain constitutive model:
    - Nonlinear elastic Eurocode2 compressive model with CEB tension stiffening (MATERIAL.CONCRETE.EUROCEB);
    - Nonlinear elastic NBR (i.e. parabolic-rectangular) compressive model with CEB tension stiffening (MATERIAL.CONCRETE.NBRCEB);
    - Mazars (1986) damage model (MATERIAL.CONCRETE.MAZARS);
    - Mazars et al. (2015) μ-Model (MATERIAL.CONCRETE.MU-MODEL);
    - Lee and Fenves (1998) plastic damage model (MATERIAL.CONCRETE.LEE.FENVES).
- Viscoelastic (MATERIAL.VISCOELASTIC) – linear viscoelastic isotropic material with constant Poisson's ratio and relaxation modulus given by a Prony series: $E(t) = E_\infty + \sum_{i=1}^{n} E_i e^{-t/\rho_i}$.
- Elastoplastic 1D constitutive models:
    - Elastoplastic material with linear isotropic and kinematic hardening (MATERIAL.ELASTOPLASTIC);
    - Elastoplastic material with exponential isotropic hardening (MATERIAL.ELASTOPLASTIC.ISOEXPHARD).
- Elastoplastic 2D/3D constitutive models:
    - Elastoplastic material (J2) with linear isotropic hardening (MATERIAL.ELASTOPLASTIC.ISOLINHARD).
- Orthotropic materials (e.g. fiber reinforced composites) with progressive failure:
    - Hashin model (MATERIAL.PROGFAIL.HASHIN);
    - Engelstad model (MATERIAL.PROGFAIL.ENGELSTAD);
    - Tsai model (MATERIAL.PROGFAIL.TSAI);
- Viscoelastic (MATERIAL.VISCOELASTIC) – linear viscoelastic isotropic material with constant Poisson's ratio and relaxation modulus given by a Prony series: $E(t) = E_\infty + \sum_{i=1}^{n} E_i e^{-t/\rho_i}$.

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The keywords and associated data are:

%MATERIAL

[#_of_materials]

%MATERIAL.CONCRETE.EUROCEB

[#_of_EUROCEB_materials]

[material_id] <f_{cm}> <ε_{c1}> <E_{cm}> <f_{ct}> <E_{ci}> <ε_y> <E_s> <$A_s/A_{c,ef}$>

%MATERIAL.CONCRETE.LEE.FENVES

[#_of_Mazars_materials]

[material_id] <E> <f_{b0}> <fc_0> <ft_0> <a_c> <b_c> <d_c> <a_t> <b_t> <d_t> <G_t> <G_c> <l_c> <l_t> <s_0>

%MATERIAL.CONCRETE.MAZARS

[#_of_Mazars_materials]

[material_id] <E> <A_c> <B_c> <A_t> <B_t> <ε_0> <Nu>

%MATERIAL.CONCRETE.MU-MODEL

[#_of_Mu-Model_materials]

[material_id] <E> <A_c> <B_c> <A_t> <B_t> <ε_{dt0}> <ε_{dc0}> <Nu>

%MATERIAL.CONCRETE.NBRCEB

[#_of_NBRCEB_materials]

[material_id] <f_c> <f_{ct}> <E_{ci}> <ε_y> <E_s> <$A_s/A_{c,ef}$>

%MATERIAL.ELASTOPLASTIC

[#_of_elastoplastic_materials]

[material_id] <E> <Nu> <f_{y0}> <K> <H>

Note: f_{y0} = initial yield stress, K = isotropic plastic modulus, and H = kinematic plastic modulus.

%MATERIAL.ELASTOPLASTIC.ISOLINHARD

[#_of_elastoplastic_linear_isotropic_hardening_materials]

[material_id] <E> <Nu> <f_{y0}> <H>

Note: f_{y0} = initial yield stress, and H = isotropic plastic modulus.

%MATERIAL.ELASTOPLASTIC.ISOEXPHARD

[#_of_elastoplastic_exponential_isotropic_hardening_materials]

[material_id] <E> <Nu> <f_{y0}> <Mu>

Note: $f_y = f_{y0}$*exp(Mu*alpha), where alpha is the equivalent plastic strain.

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%MATERIAL.ISOTROPIC

[#_of_elastic_isotropic_materials]

[material_id] <E> <Nu>

%MATERIAL.NONLINEAR.ELASTIC

[#_of_nonlinear_elastic_materials]

[material_id] <Nu> [#_of_points_of_stress_strain_curve]

<strain> <stress>

%MATERIAL.ORTHOTROPIC

[#_of_elastic_orthotropic_materials]

[material id] <E1> < E2> <E3> <Nu12> <Nu13> <Nu23> <G12> <G13> <G23>

%MATERIAL.PROGFAIL.ENGELSTAD

[#_of_Engelstad_materials]

[material id] <E1> <E2> <E3> <Nu12> <Nu13> <Nu23> <G12> <G13> <G23> <Xt> <Xc> <Yt> <Yc> <Zt> <Zc> <S12> <S13> <S23> <Alpha>

%MATERIAL.PROGFAIL.HASHIN

[#_of_Hashin_materials]

[material id] <E1> <E2> <E3> <Nu12> <Nu13> <Nu23> <G12> <G13> <G23> <Xt> <Xc> <Yt> <Yc> <Zt> <Zc> <S12> <S13> <S23> <Alpha>

%MATERIAL.PROGFAIL.TSAI

[#_of_Tsai_materials]

[material id] <E1> <E2> <E3> <Nu12> <Nu13> <Nu23> <G12> <G13> <G23> <Xt> <Xc> <Yt> <Yc> <Zt> <Zc> <S12> <S13> <S23> <Alpha>

%MATERIAL.VISCOELASTIC

[#_of_viscoelastic_materials]

[material_id] <E_∞> <Nu> [#_of_terms_of_the_Prony_series]

<E_i> <ρ_i>

%MATERIAL.DENSITY

[#_of_materials]

[material_id] <density>

Note: The material density (specific mass) is used in the computation of the mass matrix. The default value is 0.0.

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%MATERIAL.THERMAL.EXPANSION
[#_of_materials]
[material_id] <α>
Note: α = coefficient of linear thermal expansion. The default value is 0.0.

2.4.2. Sections

In the FAST program, section is an entity containing information about the material and geometry of the element cross-section. All elements (truss/frame, plate/shell, 2D and 3D solids) in FAST must have an associated section. Sections are defined by their numerical "id" and should be defined sequentially in the ascending order (1, 2, 3, …). The keywords and associated data are:

%SECTION
[#_of_sections]

%SECTION.BAR.CIRCLE
[#_of_bar_circle_sections]
[section_id] [section_material] <section_radius>

%SECTION.BAR.GENERAL
[#_of_bar_general_sections]
[section_id] [section_material] <Ax><Ay><Az><Jt><Iy><Iz><Cw><Hy><Hz>

%SECTION.BAR.INTEGRATION
'integration_method'
[#_int_points_y] [#_int_points_z]
Note: The methods available are: 'fiber' (default) and 'Gauss'. This option can be used for rectangular sections.

%SECTION.BAR.RECTANGLE
[#_of_bar_rectangle_sections]
[section_id] [section_material] <width><height>

%SECTION.BAR.TUBE
[#_of_bar_tube_sections]
[section_id] [section_material] <outer_radius> <thickness>

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% SECTION.RC.RECTANGLE

[#_of_laminated_RC_retangle_sections]

[section_id] [concrete_material] <width> <height> [#_rebars]

[rebar_material] <rebar_x_coord> <rebar_y_coord> <rebar_area>

%SECTION.FRAME3D.COUPLED

[#_of_bar_general_sections]

[section_id] [section_material] <Ax> <Ay> <Az> <Jt> <Iy> <Iz> <Cw> <Hy> <Hz>

<C11> <C12> <C13> <C14>

<C21> <C22> <C23> <C24>

<C31> <C32> <C33> <C34>

<C41> <C42> <C43> <C44>

%BEAM.SECTION.ORIENTATION

[#_of_beam_section_orientations]

[orientation_id] <orientation_vector.x> <orientation_vector.y> <orientation_vector.z>

%SECTION.FGM.2D

[#_of_FGM_2D_sections]

[section_id] [section_material] <section_thickness> [#_of_nodes]

[node_id] <Vc>

Note: This section can be used by plane stress elements. Vc is the volume fraction of the inclusions.

%SECTION.FGM.3D

[#_of_FGM_3D_sections]

[section_id] [section_material] <section_thickness> [#_of_nodes]

[node_id] <Vc>

Note: This section can be used by solid elements. Vc is the volume fraction of the inclusions.

%SECTION.FGM.SHELL

[#_of_FGM_shell_sections]

[section_id] [section_material] <section_thickness> <power_law_exponent> [#_of_integration_pts]

Note: This section can be used by shallow shell and Donnel shell elements. The integration points are used to integrate the section stresses and stiffness through the section thickness.

%SECTION.GENERAL.SHELL

[#_of_general_shell_sections]

[section_id]

<C11> … <C18>

…

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<C81> … <C88>

Note: This section can be used by shallow shell and Donnel shell elements. The 8x8 cross-sectional stiffness matrix is used directly in the analysis without modifications.

%SECTION.HOMOGENEOUS.ISOTROPIC.2D

[#_of_homogeneous_isotropic_2D_sections]

[section_id] [section_material] <section_thickness>

Note: This section can be used by plane stress, plane strain, axisymmetric, thick-plate, shallow shell, and Donnel shell elements.

% SECTION.HOMOGENEOUS.ISOTROPIC.3D

[#_of_homogeneous_isotropic3D_sections]

[section_id] [section_material]

Note: This section can be used by solid elements and general shell elements based on the degenerated solid concept.

% SECTION.HOMOGENEOUS.ORTHOTROPIC.2D

[#_of_homogeneous_orthotropic2D_sections]

[section_id] [section_material] <section_angle (deg)> <section_thickness>

% SECTION.HOMOGENEOUS.ORTHOTROPIC.3D

[#_of_homogeneous_orthotropic3D_sections]

[section_id] [section_material] <local_x-axis_x> < local_x-axis_y> < local_x-axis_z>

%SECTION.HSDT.FGM.PLATE

[#_of_HSDT_FGM_plate_sections]

[section_id] [section_material] <section_thickness> <power_law_exponent> 'HSDT_theory'

Note: 'HSDT_theory' can be ‘FSDT’ (Reissner-Mindlin/First-Order Shear Deformation Theory), TSDT (Reddy’s Third-Order Shear Deformation Theory), ‘ESDT’ (Exponential Shear Deformation Theory), ‘SSDT’ (Sinusoidal Shear Deformation Theory) or SHSDT (Soldatos’ Hyperbolic Shear Deformation Theory).

%SECTION.HSDT.PLATE

[#_of_HSDT_plate_sections]

[section_id] [section_material] <section_thickness> 'HSDT_theory'

Note: 'HSDT_theory' can be ‘FSDT’ (Reissner-Mindlin/First-Order Shear Deformation Theory), TSDT (Reddy’s Third-Order Shear Deformation Theory), ‘ESDT’ (Exponential Shear Deformation Theory), ‘SSDT’ (Sinusoidal Shear Deformation Theory) or SHSDT (Soldatos’ Hyperbolic Shear Deformation Theory).

% SECTION.LAMINATED.2D

[#_of_laminated_2D_sections]

[section_id] [#_of_laminas]

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[lamina_material] <lamina_thickness> <lamina_angle (deg)>
Note: This section can be used by plane stress elements.

% SECTION.LAMINATED.SHELL
[#_of_laminated_shell_sections]
[section_id] <local_x-axis_x> < local_x-axis_y> < local_x-axis_z> [#_of_laminas]
[lamina_material] <lamina_thickness> <lamina_angle (deg)>
Note: This section can be used by shallow shell and Donnel shell elements.

% SECTION.LAMINATED.SOLID
[#_of_laminated_solid_sections]
[section_id] <local_x-axis_x> <local_x-axis_y> <local_x-axis_z> [#_of_laminas]
[lamina_material] <lamina_thickness> <lamina_angle (deg)>

All sections can be used in geometrically nonlinear analysis. Most sections can be used in materially nonlinear analysis, except options SECTION.BAR.CIRCLE, SECTION.BAR.TUBE, SECTION.BAR.GENERAL, SECTION.FRAME3D.COUPLED, SECTION.HSDT.PLATE, SECTION.HSDT.FGM.PLATE, and SECTION.GENERAL.SHELL which can be used only in materially linear analysis.

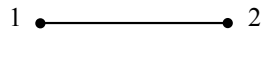
2.4.3. Elements

Elements are defined by their type, id, section, integration order (depending on the element type), and nodal incidence. It is important to note that different element types can be used in the same finite element mesh.

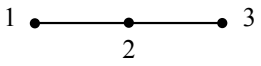
In most cases, the element type is composed by the analysis model (axisymmetric solid, plane stress, plane strain, thick plate, …) and the shape (Q4, Q8, Q9, T3, T6, …).Finite elements with geometrically nonlinear capability are indicated by the symbols TL (Total Lagrangian) or CR (co-rotational). Co-rotational elements may use only linear strain terms in the local system (PLFRAMCR1 and FRAME3DCR) or include nonlinear Green-Lagrange strains in the local system (PLFRAMCR2 and FRAME3DCRTL).

The available element shapes with the associated nodal incidences are present in Figure 1. It should be noted that the nodal incidence of plane, plate, and shell elements should always be given in counterclockwise order. Due to the use of the parametric formulation, the geometry of plane, plate, shell and solid elements may the curved in the Cartesian space.

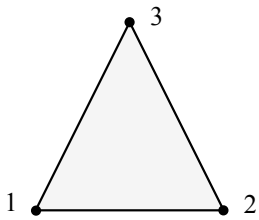
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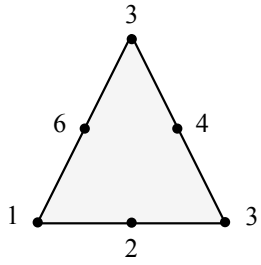
L2/TRUSS/FRAME



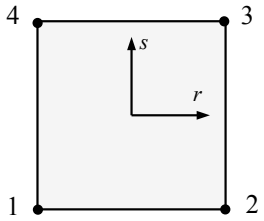
L3



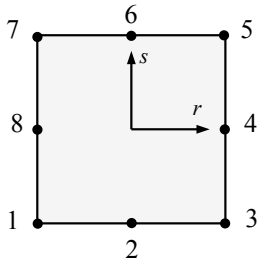
T3



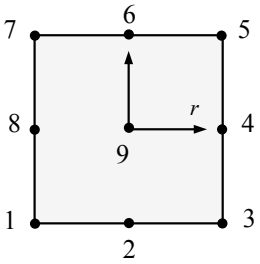
T6



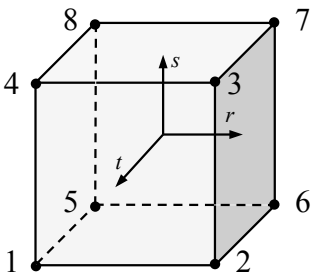
Q4



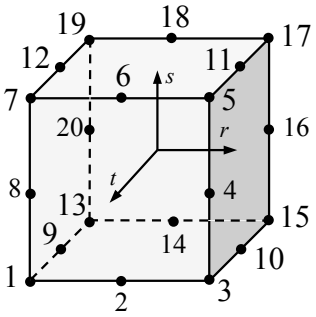
Q8



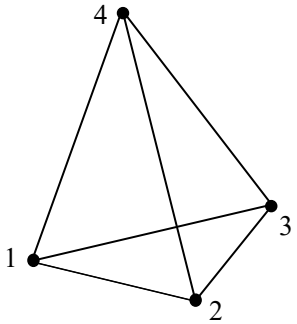
Q9



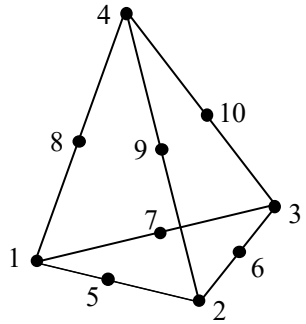
BRICK8



BRICK20



TET4



TET10

Figure 1 – Element shapes and nodal incidence.

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The element "id" should be defined sequentially in the ascending order (1, 2, 3, …). The keywords and associated data are:

```
%ELEMENT
[#_of_elements]

%ELEMENT.FORCES.OUTPUT.CONVENTION
[convention]   (1 = Direct Stiffness Method/default, 2 = Statics/Mechanics of Solids)
Note: Convention for output of nodal forces of frame/truss elements.
```

Truss elements:

```
%ELEMENT.PLTRUSS
[#_of_plane_truss_elements]
[element_id]   [element_section]   [node1]   [node2]

%ELEMENT.PLTRUSSCR
[#_of_plane_truss_co-rotational_elements]
[element_id]   [element_section]   [node1]   [node2]

%ELEMENT.PLTRUSSTL
[#_of_plane_truss_Total_Lagrangian_elements]
[element_id]   [element_section]   [node1]   [node2]

%ELEMENT.TRUSS3D
[#_of_3D_truss_elements]
[element_id]   [element_section]   [node1]   [node2]

%ELEMENT.TRUSS3DCR
[#_of_3D_truss_co-rotational_elements]
[element_id]   [element_section]   [node1]   [node2]

%ELEMENT.TRUSS3DTL
[#_of_3D_truss_Total_Lagrangian_elements]
[element_id]   [element_section]   [node1]   [node2]
```

Frame and grid elements:

```
%ELEMENT.PLFRAME
[#_of_plane_frame_elements]
[element_id]   [element_section]   [node1]   [node2]
```

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%ELEMENT.PLFRAMECR1

[#_of_plane_frame_co-rotational1_elements]

[element_id] [element_section] [node1] [node2]

Note: This co-rotational element uses linear strains in the local system.

%ELEMENT.PLFRAMECR2

[#_of_plane_frame_co-rotational2_elements]

[element_id] [element_section] [node1] [node2]

Note: This co-rotational element uses nonlinear strains (TL) in the local system.

%ELEMENT.PLFRAMETL1

[#_of_plane_frame_Total_Lagrangian1_elements]

[element_id] [element_section] [node1] [node2]

Note: This element compute the membrane strain as u,x + 1/2*v,x^2.

%ELEMENT.PLFRAMETL2

[#_of_plane_frame_Total_Lagrangian2_elements]

[element_id] [element_section] [node1] [node2]

Note:This element includes the term 1/2*u,x^2 in the membrane strains.

%ELEMENT.FRAME3D

[#_of_3D_frame_elements]

[element_id] [element_section] [section_orientation] [node1] [node2]

%ELEMENT.FRAME3DCR

[#_of_3D_frame_co-rotational_elements]

[element_id] [element_section] [section_orientation] [node1] [node2]

Note: This co-rotational element uses linear strains in the local system.

%ELEMENT.FRAME3DCRTL

[#_of_3D_frame_co-rotational_TL_elements]

[element_id] [element_section] [section_orientation] [node1] [node2]

Note: This co-rotational element uses nonlinear strains (TL) in the local system.

%ELEMENT.FRAME3DTL

[#_of_3D_frame_TotalLagrandian_elements]

[element_id] [element_section] [section_orientation] [node1] [node2]

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```
%ELEMENT.GRID
[#_of_grid_elements]
[element_id]  [element_section]  [node1]  [node2]
```

Plane stress elements:

```
%ELEMENT.PLSTRESS.BEZIER.SURFACE
[#_of_plane_stress_Bézier_surface_elements]
[element_id]  [element_section]  [integration_order]  [degree1]  [degree2] [node1] … [node_nn]
Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT.PLSTRESS.BEZIER.SURFACETL
[#_of_plane_stress_Bézier_surface_ Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [degree1]  [degree2] [node1] … [node_nn]
Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT.PLSTRESS.BEZIER.TRIANGLE
[#_of_plane_stress_Bézier_triangle_elements]
[element_id]  [element_section]  [integration_order]  [degree]  [node1] … [node_nn]
Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT.PLSTRESS.BEZIER.TRIANGLETL
[#_of_plane_stress_Bézier_triangle_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [degree]  [node1] … [node_nn]
Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT.PLSTRESS.BSPLINE.SURFACE
[#_of_plane_stress_B-Spline_elements]
[element_id]  [element_section]  [integration_order]  [patch]  [span1]  [span2]

%ELEMENT.PLSTRESS.BSPLINE.SURFACETL
[#_of_plane_stress_B-Spline_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [patch]  [span1]  [span2]

%ELEMENT.PLSTRESS.Q4
[#_of_plane_stress_Q4_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node4]

%ELEMENT.PLSTRESS.Q4TL
[#_of_plane_stress_Q4_Total_Lagrangian_elements]
```

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[element_id] [element_section] [integration_order] [node1] … [node4]

%ELEMENT.PLSTRESS.Q8
[#_of_plane_stress_Q8_elements]
[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.PLSTRESS.Q8TL
[#_of_plane_stress_Q8_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.PLSTRESS.Q9
[#_of_plane_stress_Q9_elements]
[element_id] [element_section] [integration_order] [node1] … [node9]

%ELEMENT.PLSTRESS.Q9TL
[#_of_plane_stress_Q9_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node9]

%ELEMENT.PLSTRESS.T3
[#_of_plane_stress_T3_elements]
[element_id] [element_section] [integration_order] [node1] … [node3]

%ELEMENT.PLSTRESS.T3TL
[#_of_plane_stress_T3_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node3]

%ELEMENT.PLSTRESS.T6
[#_of_plane_stress_T6_elements]
[element_id] [element_section] [integration_order] [node1] … [node6]

%ELEMENT.PLSTRESS.T6TL
[#_of_plane_stress_T6_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node6]

Plane strain elements:

%ELEMENT.PLSTRAIN.Q4
[#_of_plane_strain_Q4_elements]
[element_id] [element_section] [integration_order] [node1] … [node4]

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```
%ELEMENT.PLSTRAIN.Q4TL
[#_of_plane_strain_Q4_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node4]

%ELEMENT.PLSTRAIN.Q8
[#_of_plane_strain_Q8_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node8]

%ELEMENT.PLSTRAIN.Q8TL
[#_of_plane_strain_Q8_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node8]

%ELEMENT.PLSTRAIN.Q9
[#_of_plane_strain_Q9_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node9]

%ELEMENT.PLSTRAIN.Q9TL
[#_of_plane_strain_Q9_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node9]

%ELEMENT.PLSTRAIN.T3
[#_of_plane_strain_T3_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node3]

%ELEMENT.PLSTRAIN.T3TL
[#_of_plane_strain_T3_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node3]

%ELEMENT.PLSTRAIN.T6
[#_of_plane_strain_T6_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node6]

%ELEMENT.PLSTRAIN.T6TL
[#_of_plane_strain_T6_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node6]
```

Axisymmetric elements:

```
%ELEMENT.AXISYM.L6_INF
[#_of_axisymmetric_L6_infinite_elements_elements]
```

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[element_id] [element_section] [integration_order] [node1] … [node6]

%ELEMENT.AXISYM.Q4

[#_of_axisymmetric_Q4_elements]

[element_id] [element_section] [integration_order] [node1] … [node4]

%ELEMENT.AXISYM.Q4TL

[#_of_axisymmetric_Q4_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node4]

%ELEMENT.AXISYM.Q8

[#_of_axisymmetric_Q8_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.AXISYM.Q8TL

[#_of_axisymmetric_Q8_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.AXISYM.Q9

[#_of_axisymmetric_Q9_elements]

[element_id] [element_section] [integration_order] [node1] … [node9]

%ELEMENT.AXISYM.Q9TL

[#_of_axisymmetric_Q9_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node9]

%ELEMENT.AXISYM.T3

[#_of_axisymmetric_T3_elements]

[element_id] [element_section] [integration_order] [node1] … [node3]

%ELEMENT.AXISYM.T3TL

[#_of_axisymmetric_T3_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node3]

%ELEMENT.AXISYM.T6

[#_of_axisymmetric_T6_elements]

[element_id] [element_section] [integration_order] [node1] … [node6]

%ELEMENT.AXISYM.T6TL

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[#_of_axisymmetric_T6_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node6]

Plate elements:

%ELEMENT.HSDTPLATE.BSPLINE.SURFACE
[#_of_higher-order_plate_B-Spline_elements]
[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.HSDTPLATE.BSPLINE.SURFACETL
[#_of_higher-order_plate_B-Spline_Total_Lagrangian_ elements]
[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.THICKPLT.Q4
[#_of_Reissner-Mindlin_Q4_elements]
[element_id] [element_section] [integration_order] [node1] … [node4]

%ELEMENT.THICKPLT.Q8
[#_of_Reissner-Mindlin_Q8_elements]
[element_id] [element_section] [integration_order] [node1] … [node8]

Shell elements:

%ELEMENT.DONNELLSHELL.BSPLINE.SURFACE
[#_of_Donnell_cylindrical_shell_B-Spline_elements]
[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.DONNELLSHELL.BSPLINE.SURFACETL
[#_of_Donnell_cylindrical_shell_B-Spline_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT. DONNELLSHELL.Q8
[#_of_Donnell_cylindrical_shell_Q8_elements]
[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT. DONNELLSHELL.Q8TL
[#_of_Donnell_cylindrical_shell_Q8_Total_Lagrangian_elements]
[element_id] [element_section] [integration_order] [node1] … [node8]

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%ELEMENT. SHALLOWSHELL.BEZIER.SURFACE

[#_of_shallow_shell_Bézier_surface_elements]

[element_id] [element_section] [integration_order] [degree1] [degree2] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT. SHALLOWSHELL.BEZIER.SURFACETL

[#_of_ shallow_shell_Bézier_surface_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [degree1] [degree2] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT. SHALLOWSHELL.BEZIER.TRIANGLE

[#_of_ shallow_shell_Bézier_triangle_elements]

[element_id] [element_section] [integration_order] [degree] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT. SHALLOWSHELL.BEZIER.TRIANGLETL

[#_of_ shallow_shell_Bézier_triangle_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [degree] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT.SHALLOWSHELL.BSPLINE.SURFACE

[#_of_shallow_shell_B-Spline_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.SHALLOWSHELL.BSPLINE.SURFACETL

[#_of_shallow_shell_B-Spline_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.SHALLOWSHELL.Q8

[#_of_shallow_shell_Q8_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.SHALLOWSHELL.Q8TL

[#_of_shallow_shell_Q8_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.SHELL.BEZIER.SURFACE

[#_of_degenerated_shell_Bézier_surface_elements]

[element_id] [element_section] [integration_order] [degree1] [degree2] [node1] … [node_nn]

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Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT.SHELL.BEZIER.SURFACETL

[#_of_degenerated_shell_Bézier_surface_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [degree1] [degree2] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree1 + 1) * (degree2 + 1).

%ELEMENT.SHELL.BEZIER.TRIANGLE

[#_of_degenerated_shell_Bézier_triangle_elements]

[element_id] [element_section] [integration_order] [degree] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT.SHELL.BEZIER.TRIANGLETL

[#_of_degenerated_shell_Bézier_triangle_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [degree] [node1] … [node_nn]

Note: Number of nodes (or control points) nn = (degree + 1) * (degree+ 2) / 2.

%ELEMENT.SHELL.BSPLINE.SURFACE

[#_of_degenerated_shell_B-Spline_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.SHELL.BSPLINE.SURFACETL

[#_of_degenerated_shell_B-Spline_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2]

%ELEMENT.SHELL.Q8

[#_of_degenerated_shell_Q8_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.SHELL.Q8TL

[#_of_degenerated _Q8_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.SHELL.Q9

[#_of_degenerated_shell_Q9_elements]

[element_id] [element_section] [integration_order] [node1] … [node9]

%ELEMENT.SHELL.T6

[#_of_degenerated_shell_T6_elements]

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[element_id] [element_section] [integration_order] [node1] … [node6]

%ELEMENT.SHELL.REFERENCE.VECTOR

<reference_vector.x> < reference_vector.y> < reference_vector.z>

Note: reference vector used to define the shell local system (x', y', z'), where z' is normal to the shell midsurface. The default vector is e_y = [0, 1, 0].

%ELEMENT.SHELL.UPDATE.OPTION

'update_option'

Note: 'update_option' can be 'MR' (Moderate Rotations), 'LU' (Linear Update), 'LUN' (Linear Update and Normalization), 'QU' (Quadratic Update – default option).

Solid elements:

%ELEMENT.BRICK8

[#_of_BRICK8_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.BRICK8TL

[#_of_BRICK8_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node8]

%ELEMENT.BRICK20

[#_of_BRICK20_elements]

[element_id] [element_section] [integration_order] [node1] … [node20]

%ELEMENT.BRICK20TL

[#_of_BRICK20_Total_Lagrangian_elements]

[element_id] [element_section] [integration_order] [node1] … [node20]

%ELEMENT.BSPLINE.SOLID

[#_of_B-Spline_solid_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2] [span3]

%ELEMENT.BSPLINE.SOLIDTL

[#_of_B-Spline_solid_elements]

[element_id] [element_section] [integration_order] [patch] [span1] [span2] [span3]

%ELEMENT.TET4

[#_of_TET4_elements]

[element_id] [element_section] [integration_order] [node1] … [node4]

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```
%ELEMENT. TET4TL
[#_of_ TET4_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node4]

%ELEMENT.TET10
[#_of_TET10_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node10]

%ELEMENT. TET10TL
[#_of_ TET10_Total_Lagrangian_elements]
[element_id]  [element_section]  [integration_order]  [node1] … [node10]
```

2.5. Springs

Springs can be used to represent elastic supports or elastic connections. Elastic supports (or grounded springs) are connected to only one node, while elastic connections (i.e. flexible or semi-rigid connections) connects two different nodes.

Springs are defined by their type, id, nodal incidence, and properties.The spring properties are used todescribe the 1D force-displacement (or moment-rotation) relations.Linear springs are defined only by their stiffness, while nonlinear springs are modeled as piece-wise linear (tabular format). For displacements smaller than the first point and greater than the last point the respective forces are computed by extrapolation using the stiffness of the closest interval.

The spring "id" should be defined sequentially in the ascending order (1, 2, 3, …) for each spring type (i.e. connection and support). The keywords and associated data are:

```
%SPRING.PROPERTY
[#_of_spring_properties]
[spring_property_id]  'spring_type'     (options: 'linear' or 'nonlinear')
For 'linear' springs:
<stiffness>
For 'nonlinear' relations:
[#_points]
<displacement><force>

%SPRING.CONNECTION
[#_of_spring_connections]
```

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[spring_connection_id][node1][node2] [direction (1 to 6)] [spring_property]

%SPRING.SUPPORT
[#_of_spring_supports]
[spring_support_id][node] [direction (1 to 6)] [spring_property]

2.6. Loading

This section describes the loads applied to the finite element model. The available load types are:

- Nodal loads;
- Element loads:
    - Line loads (applied in element edges);
    - Area loads (applied in element faces);
    - Domain loads (applied in the whole element domain);
    - Frame element loads.

The load "id" should be defined sequentially in the ascending order (1, 2, 3, …) for each load type. The keywords and associated data are:

%LOAD.CASE.AREA.UNIFORM
[#_of_area_uniform_loads]
[element] [node1] [node2] [node3] [local (0 or 1)] <fx><fy><fz> (dimensions F/L^2)
Note: node1, node2, and node3 are three different vertices of the loaded surface. Loads can be given in the local (local = 1) or global systems (local = 0). The local x-axis ($\mathbf{x}_{,r}$) and y-axis ($\mathbf{x}_{,s}$) are tangent to the surface and the z-axis is normal to the surface ($\mathbf{x}_{,r} \times \mathbf{x}_{,s}$).

%LOAD.CASE.AREA.IGA.UNIFORM
[#_of_area_uniform_iga_loads]
[element] [element_face] [local (0 or 1)] <fx> <fy> <fz>

%LOAD.CASE.BODY.FORCES
[#_of_body_forces]
[element] <fx> <fy> <fz> (dimensions F/L^3)
Note: The forces fx , fy, and fz are considered constant within the element.

%LOAD.CASE.DONELL.UNIFORM
[#_of_area_uniform_Donnell_shell_loads]
[element] [cartesian_system (0 or 1)] <fx> <fy> <fz> (dimensions F/L^2)

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Note: The forces fx , fy, and fz are considered constant within the element.

%LOAD.CASE.ELEMENT.TEMPERATURE

[#_of_element_with_prescribed_temperature]

[element] <ΔT_{bottom}> <ΔT_{top}>

Note: The temperatures variations at the bottom and top surfaces are considered constant within the element.

%LOAD.CASE.FRAME3D.UNIFORM

[#_of_frame3D_forces]

[element] [local (0 or 1)] <fx> <fy> <fz> (dimensions F/L)

Note: Loads can be given in the local (local = 1) or global systems (local = 0). The local x-axis is tangent to the element axis, the y-axis is defined by the section orientation vector, and the z-axis is perpendicular for both x-axis and y-axis.

%LOAD.CASE.GRID.UNIFORM

[#_of_grid_forces]

[element] <fz> (dimension F/L)

%LOAD.CASE.NODAL.FORCE

[#_of_nodal_forces]

[node] <Fx> <Fy> <Fz> <Mx> <My> <Mz>

%LOAD.CASE.LINE.FORCE.UNIFORM

[#_of_line_uniform_loads]

[element] [node1] [node2] [local (0 or 1)] <fx> <fy> <fz> (dimensions F/L)

Note: node1 and node2 are the end nodes (vertices) of the loaded edge. Loads can be given in the local (local = 1) or global systems (local = 0). The local x-axis is tangent to the edge ($\mathbf{x}_{,r}$), the y-axisis normal to the edge, and the z-axis is out-of-plane.

%LOAD.CASE.LINE.FORCE.IGA.UNIFORM

[#_of_line_uniform_iga_loads]

[element] [element_face] [element_face_edge] [local (0 or 1)] <fx> <fy> <fz>

Note: [element_face] and [element_face_edge] are defined considering element incidence. [element_face] can be: 'Front', 'Back', 'Left', 'Right', 'Above' and 'Below'.

[element_face_edge] can be: 'Top', 'Bottom', 'Left' and 'Right'.

%LOAD.CASE.PLFRAME.UNIFORM

[#_of_plframe_forces]

[element] [local (0 or 1)] <fx> <fy> <mz> (dimensions F/L, F/L, and FL/L)

Note: Loads can be given in the local (local = 1) or global systems (local = 0). The local x-axis is

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tangent to the element axis and the y-axis is normal to the element.

%LOAD.CASE.SHELL.UNIFORM

[#_of_area_uniform_degenerated_shell_loads]

[element] [cartesian_system (0 or 1)] <fx> <fy> <fz> (dimensions F/L^2)

Note: The forces fx , fy, and fz are considered constant within the element.

For time-dependent or nonlinear static analyses, two different loading modes (HEADER.ANALYSIS.PROPORTIONAL.LOADING) can be considered: proportional or non-proportional. The use of proportional loading means that all loads are increased or decreased by the same factor. This is the default option for nonlinear static problems and should be used with most path-following methods (e.g. arc-length). On the other hand, the use of non-proportional loading allows each load to have a different time variation. This is default option for time-dependent and dynamic analyses. In addition, it can be used in nonlinear analyses static by the Newton-Raphson method of problems subjected to cyclic loads or with loading-unloading-reloading.

The available time functions are piece-wise linear, harmonic (sinusoidal) e half-sine. The keywords and associated data are:

%TIME.FUNCTION.TABLE

[#_points]

<time><value>

%TIME.FUNCTION.HARMONIC

<amplitude><period><phase_angle (rad)><start_time><end_time>

Note: time_function = amplitude*sin(2*π*t/period + phase) for start_time < t < end_time. Otherwise, time_function = 0.

%TIME.FUNCTION.HALFSINE

<amplitude><period><rest_time>

Note: This function may be used to represent a repeating pulse given by a half-sine function, time_function = amplitude*sin(2*π*t/period), followed by a rest time where time_function = 0.

For non-proportional loading all loads have an associated time function. The value of a load F at a given time t is computed from $F(t) = Q*f(t)$, where Q is the given load and $f(t)$ is the associated time function. A time function can be used for a load or a group of loads. The time function should be given in the input file **before** the associated loads. This function will

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be associated with all the following loads until a new time function appears in the input file, as describe below for an example with two load groups:

```
TIME.FUNCTION

%REMARK
Load group associated with the first time function.

LOAD
….
LOAD

TIME.FUNCTION

%REMARK
Load group associated with the second time function.

LOAD
…
LOAD
```

Loads appearing before the first time function will be considered constant (i.e. dead loads).

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3. OUPUT FILE

The output file generated by FAST contains initially a copy of the input data followed the computed results. Result information is organized in a result case. Currently FAST considers only one result case. Each result case is organized in a result case step. Each result case step may be associated with a time step (time-dependent analysis) or a load factor (nonlinear analysis). The keywords and associated data are:

```
%RESULT
[#_of_result_cases]
[result_case_id]    'result_case_title (12)'

%RESULT.CASE
[current_result_case_id]   [#_of_steps]
[step_id]    'label (12)'

%RESULT.CASE.STEP
[current_case_step_id]

%RESULT.CASE.STEP.TIME
<time_step>

%RESULT.CASE.STEP.FACTOR
<load_factor>

%RESULT.CASE.STEP.NODAL.DISPLACEMENT
[#_of_nodes]   'nodal_displacement_label (12)'
[node_id]   <dx>   <dy>  <dz>  <rx>  <ry>  <rz>

%RESULT.PATCH.DISPLACEMENT
[#_of_patch_points]
[patch_point_id]   <dx>   <dy>  <dz>  <rx>  <ry>  <rz>

%RESULT.CASE.STEP.NODAL.SCALAR
[#_of_scalar_fields]
'field_label_1 (12)'  'field_label_2 (12)'  ...  'field_label_#_of_scalar_fields(12)'
%RESULT.CASE.STEP.NODAL.SCALAR.DATA
[#_of_nodes]
[node_id]    <field_1_dat>  <field_2_dat>  ...  <field_#_of_scalar_fields_dat>
```

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%RESULT.CASE.STEP.ELEMENT.NODAL.SCALAR
[#_of_scalar_fields]
'field_label_1 (12)' 'field_label_2 (12)' ... 'field_label_#_of_scalar_fields (12)'

%RESULT.CASE.STEP.ELEMENT.NODAL.SCALAR.DATA
[#_of_elements]
[element_id] (assuming: n = number of nodes of current element)
<field_1_node_1> <field_2_node_1> ... <field_#_of_scalar_fields_node_1]
<field_1_node_2> <field_2_node_2> ... <field_#_of_scalar_fields_node_2]
:
<field_1_node_n> <field_2_node_n> ... <field_#_of_scalar_fields_node_n]

%RESULT.CASE.STEP.PATCH.SCALAR
[#_of_scalar_fields]
'field_label_1 (12)' 'field_label_2 (12)' ... 'field_label_#_of_scalar_fields (12)'

%RESULT.CASE.STEP.PATCH.SCALAR.DATA
[#_of_patch_points]
if (patch_stress_smooth is true)sss
[patch_point]
[element_id] <field_1> <field_2> … <field_#_of_scalar_fields>
else
[patch_point] [number_of_elements]
For each element:
[element_id] <field_1> <field_2> … <field_#_of_scalar_fields>

%RESULT.CASE.STEP.ELEMENT.GAUSS.SCALAR
[#_of_scalar_fieds]
'field_label_1 (12)' 'field_label_2 (12)' ... 'field_label_#_of_scalar_fields (12)'

%RESULT.CASE.STEP.ELEMENT.GAUSS.SCALAR.DATA
[#_of_elements]
[element_id] [#_of_Gauss_points] (assuming: m = number of Gauss points)
<field_1_Gauss_1> <field_2_Gauss_1> ... <field_#_of_scalar_fields_Gauss_1>
<field_1_Gauss_2> <field_2_Gauss_2> ... <field_#_of_scalar_fields_Gauss_2>
:
<field_1_Gauss_m> <field_2_Gauss_m> ... <field_#_of_scalar_fields_Gauss_m>

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